

Origami-Inspired Ion-Propelled Microrobots With High Thrust-to-Weight Ratio and Rapid Assembly

Xianfa Cai , Member, IEEE, Yuwang Shen, Yingxu Tian, Yunqi Cao , Senior Member, IEEE, Qiannan Tao , Member, IEEE, and Wei Li , Senior Member, IEEE

Abstract—Ion-propelled flying microrobots utilize electrohydrodynamic (EHD) thrust to generate airflow without moving parts. However, existing platforms often struggle to balance thrust-to-weight ratio, fabrication cost, and assembly complexity. To address this, this study presents an origami-inspired ion-propelled flying microrobot (IFM) fabricated from a single-piece carbon fiber–polyimide laminate. The design allows for rapid manual folding and snap-fit locking in under 1 min. By minimizing adhesive use, it effectively reduces mass overhead and improves the repeatability of the emitter–collector gap. For propulsion, the IFM employs a corona-discharge configuration featuring an emitter electrode suspended above a gridded collector electrode. Two geometries (triangular and square) are developed and characterized in terms of current, wind speed, and thrust, supported by multiphysics simulation for collector design selection. The square IFM achieves tethered vertical takeoff with a peak thrust-to-weight ratio of 3.14 under laboratory conditions. A derived EHD model linking corona onset, space-charge current, thrust scaling, and thrust-per-power trends justifies the electrode gap and collector grid parameters. Proportional–integral–derivative (PID) altitude regulation is demonstrated in a tethered configuration using external motion capture feedback and fast emitter-voltage

modulation superimposed on a high-voltage bias. This research highlights the potential of origami-inspired techniques in enabling complex structural designs for micro-robotic flight systems.

Index Terms—Microrobot, origami, ion propulsion, multiphysics simulation, carbon fiber, efficient assembly, altitude control, thrust-to-weight ratio.

I. INTRODUCTION

FLYING microrobots have emerged as a critical research area due to their versatility in applications, such as disaster rescue, surveillance, and planetary exploration [1], [2]. Compared to larger aircraft, flying microrobots offer unique advantages in maneuverability and operational flexibility, particularly in confined environments (e.g., smaller turning radius and less clearance required near obstacles). However, challenges such as propulsion miniaturization, limited load-carrying capacity, and manufacturing complexity remain significant hurdles [3], [4].

Among flying microrobots, ion-propelled flying microrobots (IFMs) stand out because they leverage the electrohydrodynamic (EHD) effect to generate thrust without mechanical moving parts [5], [6], [7]. A voltage difference between the emitter and collector ionizes nearby air molecules; the generated ions transfer momentum to neutral molecules, producing ion wind and propulsion [see Fig. 1(a)]. This thrust can be modulated by the applied voltage. Compared with flapping- or rotary-wing systems, EHD operates quietly, requires no moving parts, and is suitable for applications needing stealth [3], formation keeping [8], or mechanically simpler designs [9].

State-of-the-art IFMs are typically fabricated using silicon-based materials (e.g., SOI processes) or micro-machined metal electrodes, with structural gaps precisely controlled in the range of 3.5–30 mm [6], [7], [10]. For instance, silicon electrode-based IFMs can weigh as little as 2.5–37 mg, with four-thruster configurations achieving thrust-to-weight ratios exceeding 1.5 in sizes ranging from 1.8 to 2.5 cm [7], [11]. Recent advancements in laser processing technology have enabled the use of stainless steel emitters and collector grids, achieving thrust-to-weight ratios of approximately 3 at a weight of 37 mg [6]. However, thrust loss caused by factors such as the collector grid's geometry remains a challenge [7]. Concurrently, silicon and metallic

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Xianfa Cai, Yuwang Shen, Yingxu Tian, and Qiannan Tao are with the College of Integrated Circuit Science and Engineering, Nanjing University of Posts and Telecommunications, Nanjing 210046, China (e-mail: xianfacai@njupt.edu.cn; shenyuwang1217@163.com; ty-yingxu@163.com; qntao_lea@njupt.edu.cn).

Yunqi Cao is with the College of Control Science and Engineering, Zhejiang University, Hangzhou 310027, China (e-mail: caoyunqi@zju.edu.cn).

Wei Li is with the College of Integrated Circuit Science and Engineering, Nanjing University of Posts and Telecommunications, Nanjing 210046, China, and also with the School of Mechanical Science and Engineering, Huazhong University of Science and Technology, Wuhan 430074, China (e-mail: wei.li@njupt.edu.cn).

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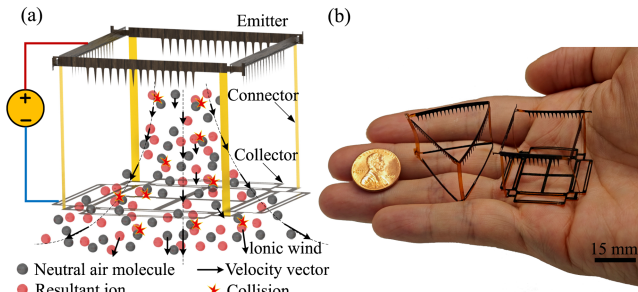


Fig. 1. Overview of origami-inspired IFMs. (a) Schematic illustration of the flying principle. (b) Triangular and square geometric prototypes held in hand, shown alongside a coin for scale.

thin-film materials require advanced equipment, cleanroom facilities, and prolonged manufacturing procedures, which considerably increase fabrication cost [12]. In addition, the overall thrust-to-weight ratio typically ranges from 1 to approximately 3.0, leaving a limited margin for stable flight with miniature sensors [6]. Furthermore, the limited dynamic response of conventional high-voltage (HV) power systems restricts rapid thrust tracking and introduces latency into altitude-control commands. Consequently, stable closed-loop control for IFMs remains challenging, and current efforts mainly enhance passive stability by reducing disturbance sensitivity through structural optimization [13], [14].

Recent advances in soft aerial robotics have demonstrated compliant mechanisms for lightweight and deformable flight systems [15], [16], [17]. Drawing on the structural advantages of origami, this study presents an ion-propelled flying micro-robot fabricated from a single carbon fiber–polyimide (CF–PI) laminate. In this architecture, the carbon fiber layer functions as both a lightweight structural frame with high specific stiffness and a conductive electrode, whereas the polyimide layers form compliant hinges, provide localized electrical insulation, and generate elastic preload for snap-fit locking. Unlike soft-bodied or actively morphing aerial platforms, the proposed design localizes compliance to the folding and locking regions rather than using structural deformation for in-flight shape adaptation. The remaining regions retain sufficient stiffness to maintain a stable emitter–collector gap, which is essential for reliable corona discharge and repeatable thrust generation. Compared with our previous ultralight aluminized-film IFMs and multi-engine ion-wind microrobots [18], [19], the proposed design adopts a more integrated single-laminate architecture. The 2-D planar laminate can be rapidly folded and mechanically locked into a lightweight 3-D propulsion structure without separate stiffeners or extensive adhesive bonding. Because the carbon fiber simultaneously performs structural and electrical functions, no additional metal-plating layer is required. This single-piece construction reduces parasitic mass and assembly complexity while improving discharge tolerance and device-to-device consistency. Furthermore, unlike conventional multilayer pop-up assemblies that rely on separate structural components and adhesive layers [20], [21], the proposed IFM uses geometric stops and snap-fit features to define the assembled configuration and maintain the emitter–collector spacing with minimal

manual adjustment. This structural design also simplifies the manufacturing workflow. A batch of tens of planar devices can be fabricated in approximately 1.5 h, after which each device can be folded and secured manually using only tweezers in less than one minute. In comparison, conventional multistep silicon micromachining generally requires prolonged operation using specialized microfabrication equipment in a cleanroom [6], [7], [10]. The proposed batch-oriented process therefore reduces the per-device fabrication and assembly time while avoiding extensive cleanroom processing. As depicted in Fig. 1(b), triangular and square origami-based IFMs were fabricated. The triangular IFM consists of three emitter and collector sides, whereas the square IFM consists of four. These geometries represent simple fully enclosed polygonal configurations that can generate symmetric thrust and can both be formed from a single 2-D planar laminate.

Beyond simplifying fabrication and assembly, the single-laminate strategy also facilitates the coordinated design of the structure and EHD propulsion system. Accordingly, this study adopts a design-oriented modeling framework in which the electrode gap, collector open-area ratio, and flow-focusing geometry are jointly considered to improve the effective drift area and stable discharge current while maintaining an adequate breakdown margin. In addition, the single-piece CF–PI platform is integrated with tethered closed-loop altitude regulation using external motion-capture feedback, proportional–integral–derivative (PID) control, and fast emitter-voltage modulation.

To the best of the authors' knowledge, this study represents the first attempt to employ origami techniques in the fabrication of IFMs while demonstrating tethered closed-loop altitude regulation with external sensing. This approach results in lightweight, flexible, high-strength, and low-cost designs with high performance. This article details the IFM's design, fabrication, and performance, concluding with future research directions.

II. MATERIAL AND METHODS

A. Structure and Assembly

Inspired by origami techniques, the IFM design uses conductive CF materials known for their high strength and lightweight properties as the main components of the emitter and collector electrodes. To enhance structural continuity and provide compliant folding regions, PI films were laminated onto both sides of the CF layer.

As shown in Fig. 2(a), the laminate material used for the origami IFMs is a composite consisting of three layers: a central layer of CF sandwiched between two PI films of varying thicknesses. After laminating, the composite material is laser-cut into the desired shapes for IFM assembly. The triangular IFM is assembled using a snap-fit mechanism located at each end of its chain-like structure [see Fig. 2(b)]. This snap-fit mechanism significantly simplifies the assembly process, enabling completion within just 1 min using only tweezers. In most cases, the assembly relies solely on mechanical connections and rarely requires additional adhesives. This feature is crucial in maintaining the lightweight nature of the structure.

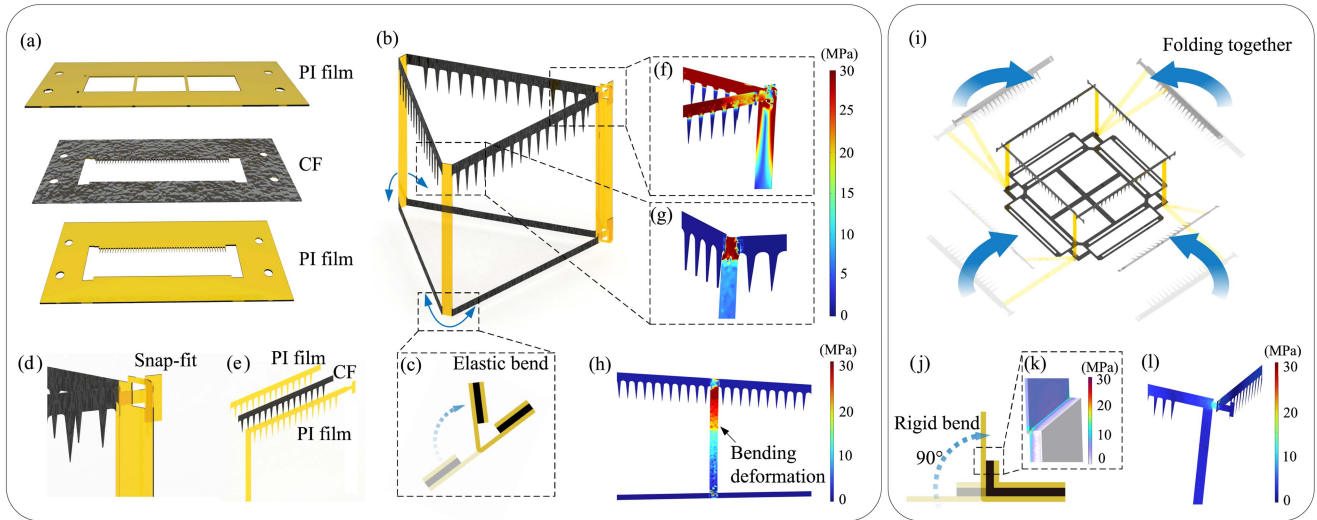


Fig. 2. Assembly methods of the triangular and square IFMs. (a) Structure of the laminate material. (b) Chain closed-loop assembly form of the triangular IFM. (c) Bending process at the elastic bend portion. (d) Structure of the snap-fit. (e) Stacking structure of the emitter. (f) FEM analysis of the mechanical behavior of the snap-fit. (g) FEM analysis of the mechanical behavior of 20 μm -thick connectors. (h) FEM analysis of the mechanical behavior of 20 μm -thick connectors. (i) Folding assembly mechanism of the square IFM. (j) Bending process at the rigid-bending region. (k) FEM analysis of the mechanical behavior at the rigid-bending portion. (l) Stress distribution at the snap-fit, where the lateral load disturbance is three times the gravity.

The triangular IFM is constructed through elastic bending segments and snap-fit connections [see Fig. 2(c) and (d)]. These connections deliver geometric constraints and preload through PI's elasticity. The stacking structure of the emitter after assembly is shown in Fig. 2(e). This three-layer structure ensures both excellent electrical conductivity and structural integrity. Although assembly is manual, the snap-fit mechanism relies on precisely manufactured physical stops, which primarily define the structure's geometry. This geometric constraint also reduced device-to-device gap variation, with emitter-collector gap standard deviations of 0.26 mm for the square IFMs and 0.24 mm for the triangular IFMs across five assembled devices of each geometry.

Finite element models (FEM) are developed using COMSOL Multiphysics to simulate the mechanical behavior of the snap-fit and connectors after assembly [see Fig. 2(f) and (g)]. The elastic-bending assembly results indicate that the interfaces between the snap-fit and the connector joints are subjected to relatively high stress. The maximum stress of 30 MPa is significantly lower than the critical bending collapse stress of 600 MPa for CF. As a result, the postassembly structure retains its ideal shape without significant deformation or collapse, demonstrating the effectiveness of the origami techniques and the careful design of the materials, dimensions, and thicknesses.

However, if the PI film is too thin, then it may lead to structural collapse. For instance, as shown in Fig. 2(h), using a 20 μm PI film instead of the intended 25 μm design results in significant bending deformation at the connector's center. A critical consideration in the overall structural design is ensuring that the connector, weighing approximately 5 mg and being the component most susceptible to collapse, can support the emitter's weight without experiencing significant deformation.

The assembly process for the square IFM differs from that of the triangular IFM. It employs a folding assembly mechanism in

which the four connectors are rigidly bent at 90° angles before being secured with a snap-fit mechanism [see Fig. 2(i)]. The rigid bending process is illustrated in Fig. 2(j). The presence of CF reduces the stress at the rigid bending portions, with only minor stress (maximum 20 MPa) observed at the connection points between the connectors and the rigid bending sections [see Fig. 2(k)].

During the takeoff process, the connectors undergo a transition from compressive stress to tensile stress. Under transverse load disturbances of three times the gravity, the maximum stress at the snap-fit's location is only 30 MPa. This results in insignificant deformations as depicted in Fig. 2(l). The maximum thrust is in the mN range; therefore, even multigravitational disturbances correspond to inertial loads of mN scale for a robot weighing less than 0.1 g. These loads are several orders of magnitude smaller than the stress margin inferred from the static assembly stress analysis. Consequently, the stress concentration driven by the assembly remains the key structural design constraint in the current tethered experiments.

The rigid bending assembly method improves the stability of the square IFM, although it requires additional heat molding. Each emitter uses equidistant tip arrays to concentrate the electric field and initiate corona discharge. Within the limits of laser-cutting precision, minimizing the tip angle is important for enhancing discharge performance.

B. Material Processing Technique

Fig. 3 provides a clear visualization of the IFM planar structure fabrication process. First, each layer is fabricated individually via a laser cutting instrument (SM-F10, Seaman Inc.) that utilizes an ultraviolet cold light source to reduce the risk of thermal damage. Next, a locating-pin tool is employed to align the layers, followed by preheating at a temperature of 140 °C for 5 min. This step facilitates the adhesion of PI onto the CF

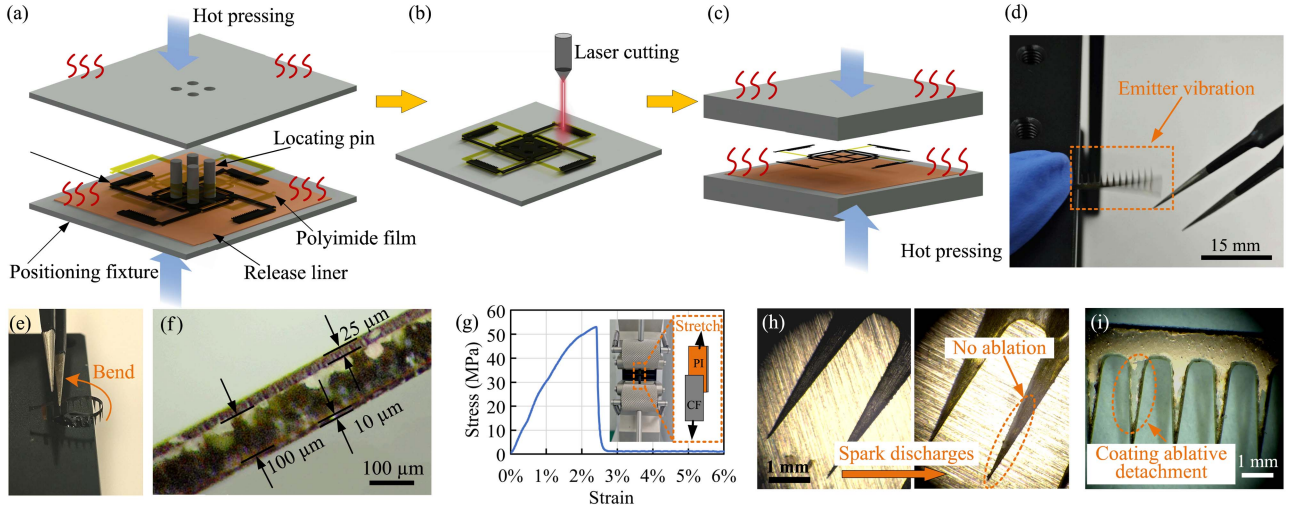


Fig. 3. Manufacturing of the IFM and its material characteristics. (a) High-temperature prelamination of the laser-cut films. (b) Laser cutting of the prelaminated IFM. (c) High temperature lamination of the flat IFM to be assembled. (d) Photograph showing the bending vibration of the emitter. (e) The emitter flips 180°. (f) Cross-section of CF layer laminated between two PI films. (g) Shear experiment results on CF and PI composite materials, including photographs of the experimental setup and a schematic diagram of the shear testing process. (h) Discharge damage test on the emitter made from CF composites. (i) Discharge damage test on the emitter made from aluminized film for comparison.

pregreg, as demonstrated in Fig. 3(a). Subsequently, as shown in Fig. 3(b), a release cut is performed. Finally, the lamination process is carried out at a regulated temperature of 140 °C using a pressure of 5 kPa for 1 h. This ensures the effective curing of the adhesive, as depicted in Fig. 3(c).

After CF-PI lamination, the composite shows advantages over typical silicon and aluminum structures, including robustness, flexibility, and elasticity [14], as shown in Fig. 3(d). The emitter fully recovers after a terminal flip of 180° [see Fig. 3(e)]. The sandwich laminate of CF and two PI layers exhibits tight adhesion, high uniformity, and no visible bubbles or wrinkles [see Fig. 3(f)]. The shear test in Fig. 3(g) shows a shear strength of 52.8 MPa for a 3 mm overlap between CF and PI, which is much higher than the assembly-induced stress.

During corona discharge, CF conductivity avoids additional plating. Prolonged HV operation can oxidize CF surfaces, but PI encapsulation reduces damage. The laminate shows no ablation after 50 destructive spark discharges [see Fig. 3(h)], whereas metal-plated films can delaminate [see Fig. 3(i)] due to Joule heating and breakdown ablation. This demonstrates the superior durability of CF-based materials in discharge scenarios.

Another key advantage of CF-PI laminated films is their low density (1.5 g/cm³), which is lower than that of silicon or metal-plated materials (2–4 g/cm³) and improves the thrust-to-weight ratio for integrating functional sensors and chips. The mechanical, electrical, and thermal properties of these CF-PI composite materials reflect their ability to withstand the stresses and high-voltage corona environments encountered during IFM operation.

III. STRUCTURAL DESIGN OF THE IFM

A. Dimension Parameters

Fig. 4(a) and (b) show the dimensional designs for triangular and square IFMs, with identical emitter tip and connector

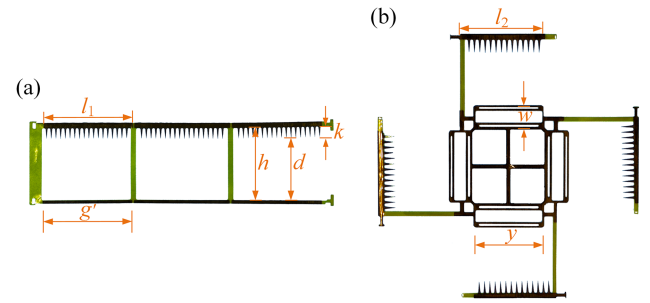


Fig. 4. Dimensional design of the IFMs. (a) Triangular IFM dimensions. (b) Square IFM dimensions.

TABLE I
STRUCTURE PARAMETERS

Parameter	Value
Length of triangular IFM emitter l_1	29 mm
Length of square IFM emitter l_2	24 mm
Length of tip k	3.7 mm
Connector height h	21.7 mm
Distance between tip and collector $d = h - k$	18 mm
Width of triangular IFM collector g'	29 mm
Width of collector sub-unit w	8 mm
Length of collector sub-unit y	17.6 mm
Angle of tips β	10°

parameters as detailed in Table I. The triangular emitter has a length $l_1 = 29$ mm, while the square emitter has $l_2 = 24$ mm. Both designs feature a row of tips at the bottom, with a fixed spacing and tip length $k = 3.7$ mm, which is conducive to stable ion wind generation.

While smaller tip angles enhance ion wind efficiency by concentrating the electric field more effectively, the emitter tip angle is set at 10° to balance material charring during CF cutting and laser cutting precision. The connector height h is 21.7 mm, and the distance d between the collector and the emitter tips is

calculated as $d = h - k = 18$ mm. The collector grid bars have a uniform linewidth of 0.45 mm, determined by laser cutting precision, film alignment accuracy, and deformation during hot pressing.

B. Modeling of EHD Thrust

EHD thrust arises from the movement of a fluid carrying diffused charged particles under the influence of an external electric field. For the tested electrode gap and stable-corona voltage range, the following model is used to describe the main thrust trend. The corona discharge is treated as quasi-steady, and the induced airflow is considered low-speed and laminar before spark breakdown occurs. Under drift-dominated charge transport, the current can be expressed as [18], [22]

$$I = \int_A \rho_e \mu \mathbf{E} \cdot \mathbf{n} dA \quad (1)$$

where A is a cross-sectional surface with outward normal $\mathbf{n}$, ρ_e is the charge density, $\mathbf{E}$ is the electric field, and μ is the ion mobility in air. In this study, ion mobility is assumed to be $2 \times 10^{-4} \text{ m}^2/(\text{V} \cdot \text{s})$.

The Coulomb body force density acting on neutral flow is $\mathbf{f} = \rho_e \mathbf{E}$, and the net thrust can be written as

$$F = \int_{\Omega} \rho_e \mathbf{E} \cdot \mathbf{e}_z d\Omega \quad (2)$$

where Ω denotes the interelectrode volume and $\mathbf{e}_z$ is the thrust direction. Combining (1) and (2) yields a widely used thrust identity

$$F = \int_0^d \frac{I}{\mu} dx = \frac{Id}{\mu}. \quad (3)$$

Equation (3) offers clear implications for the design process. At a fixed gap d , any increase in thrust requires a higher current I while maintaining stable discharge. This equation is therefore used here to describe the dependence of EHD thrust on the discharge current and electrode gap during geometry selection.

To determine the resulting operating window, we can estimate the corona inception voltage using a wire-to-plane approximation based on Peek-type models [11]

$$V_0 = m_v g_0 R \ln\left(\frac{d}{R}\right) \quad (4)$$

where m_v depends on surface condition, g_0 is the breakdown field strength of air, and R is the tip curvature radius. In practice, V is chosen above V_0 but below the breakdown-limited voltage V_B . Near V_B , discharge intermittency increases and thrust ripple becomes larger, which affects altitude regulation accuracy.

To describe the space-charge-limited current and thrust scaling, assuming a space-charge-limited regime in the drift region, the current density follows the Mott–Gurney law [23]

$$j = \frac{9}{8} \mu \varepsilon_0 \frac{V^2}{d^3} \quad (5)$$

where ε_0 is the vacuum permittivity. Accounting for the corona onset requires a shifted-voltage form

$$I = S_{\text{eff}} j = S_{\text{eff}} \frac{9}{8} \mu \varepsilon_0 \frac{(V - V_0)^2}{d^3} \quad (6)$$

where S_{eff} is the effective cross-sectional area available for ion drift, which is influenced by the collector open-area ratio ϕ . Here, $V - V_0$ denotes the voltage above corona onset. This equation shows that the discharge current increases with S_{eff} and decreases with the electrode gap d . It is therefore used to compare different collector openings and electrode gaps during geometry selection.

Substituting (6) into (3) yields a direct thrust scaling law

$$F = \frac{9}{8} \varepsilon_0 S_{\text{eff}} \frac{(V - V_0)^2}{d^2}. \quad (7)$$

Equation (7) suggests that reducing the gap d increases the thrust approximately as $1/d^2$ for a fixed $(V - V_0)$. However, because the maximum stable voltage is constrained by breakdown, a practical engineering limit is given by

$$V \leq V_B = E_B d \quad (8)$$

where E_B is the effective breakdown field, which is influenced by humidity, tip curvature, and surface contamination [10]. This explains why simply increasing the voltage is not a scalable strategy, since the margin for stable operation diminishes as V approaches the breakdown threshold V_B , even though thrust scales positively with V .

We relate the collector design to S_{eff} as follows:

$$S_{\text{eff}} = \kappa(\phi, w) \phi S_{\text{geo}} \quad (9)$$

where S_{geo} is the geometric drift cross-section, $\phi \in (0, 1)$ is the collector open-area ratio, and $\kappa(\phi, w)$ is a focusing factor describing how grid spacing and subunit width w concentrate the ion wind.

Combining (4)–(8) establishes a breakdown-limited thrust trend. Substituting $V = V_B = E_B d$ into (7) yields

$$F_{\text{max}}(d) = \frac{9}{8} \varepsilon_0 S_{\text{eff}} \left(E_B - \frac{V_0(d)}{d} \right)^2 \quad (10)$$

where $V_0(d)$ is defined in (4). Equation (10) reveals the combined influence of the breakdown limit and the gap-dependent corona onset voltage on thrust. Consequently, practical engineering dictates selecting d to ensure V_0 remains significantly lower than the maximum supply voltage and operation near V_B is avoided during control transients to mitigate discharge intermittency. This rationale justifies our selection of an 18 mm gap, which provides stable operation and an adequate thrust margin while maintaining frame compactness.

For a collector with periodic bars of width b and a center-to-center pitch p , the open-area ratio can be approximated as

$$\phi = 1 - \frac{b}{p}. \quad (11)$$

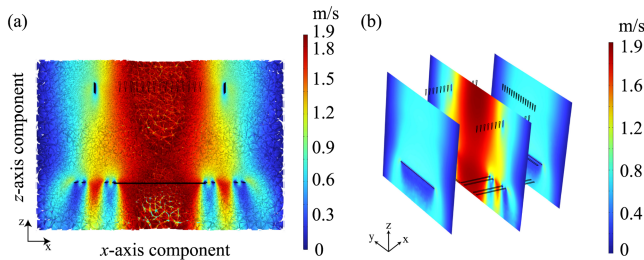


Fig. 5. 2-D cross-sectional view of simulated wind speed fields around the IFM. (a) Wind speed distributions of the collector center cross-section. (b) Comparison of wind speed distributions between collector center cross-section and edge cross-section.

Equation (11) is convenient for fabrication-aware design because b is often bounded by laser-cut linewidth and alignment tolerance. Decreasing p improves electrostatic focusing and can increase κ , but it reduces ϕ and increases pressure loss; therefore, p should be selected jointly with w to maximize S_{eff} in (9) rather than maximizing either openness alone.

IV. SIMULATION ANALYSIS

FEM simulations using COMSOL Multiphysics were conducted to optimize the collector width w of the square IFM, as w directly dictates the collector's focusing efficiency and the effective current collection area [19], [24]. The simulations integrated electrostatic fields, transport of diluted species, and laminar flow dynamics. Species-transport boundary conditions were specified as follows: a portion of the needle tip served as a species inlet at a prescribed concentration, and the collectors were assigned as outflow boundaries. For the fluid flow, the top boundary of the domain was set as a static-pressure inlet, the bottom boundary as a static-pressure outlet, and the lateral boundaries as no-slip walls. The emitter was modeled as an electrode biased at 11.09 kV, and the collector as a grounded grid. The fluid domain utilized standard atmospheric air properties, including a density of 1.2 kg/m^3 , a relative permittivity of 1, a dynamic viscosity of $1.8 \times 10^{-5} \text{ Pa} \cdot \text{s}$, and a thermal conductivity of $25 \times 10^{-3} \text{ W/(m} \cdot \text{K)}$. Furthermore, the ion diffusion coefficient was set to $5.3 \times 10^{-5} \text{ m}^2/\text{s}$. A mesh-independence study was performed by reducing the minimum element size near the emitter tips and collector-grid edges from 70 to $35 \mu\text{m}$, which changed the predicted peak wind speed by less than 1.0%, confirming adequate resolution.

Fig. 5(a) and (b) reveal that airflow velocity peaks in the center and drops near the edges, strongly supporting a collector design tailored for central focusing.

The results shown in Fig. 6 highlight wind speeds ranging from 1.50 to 1.90 m/s, with maximum speeds concentrated near the center of the cylinder. While additional grid bars enhance wind speed by converging airflow, excessive bars result in pressure loss and reduced average wind speed.

To optimize w , simulations were conducted for widths ranging from 6 to 10 mm. A collector width of 8 mm achieved

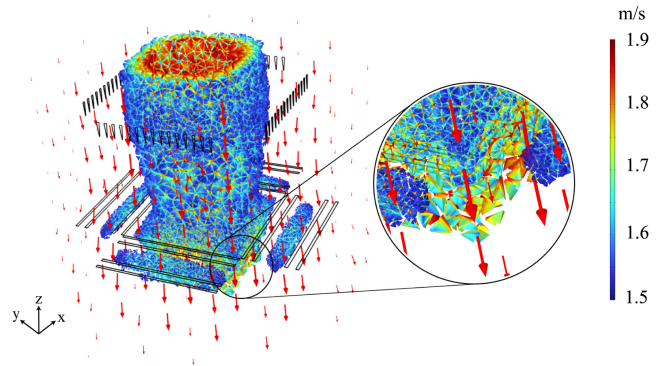


Fig. 6. 3-D visualization of simulated wind speed for the IFM.

the highest wind speed of 1.90 m/s, reflecting the fundamental tradeoff in (9) between flow focusing and aerodynamic blockage.

V. EXPERIMENTAL CHARACTERIZATION AND TETHERED FLIGHT

A. Collector Grid Spacing Validation

The square IFM collector incorporates auxiliary grid bars to improve flow focusing. As implied by (9), adding grid bars may increase focusing but can reduce openness and introduce aerodynamic blockage. To validate the tradeoff experimentally, the spacing of the auxiliary grid bars was varied from 3 to 7 mm (1 mm increments) while keeping other parameters unchanged. Experiments were conducted under positive corona in an indoor environment at $21 \pm 0.5^\circ\text{C}$ and relative humidity of 43%.

The IFMs were powered by a voltage source (DW-P503-4ACD2, Dongwen Inc.), and the supply current was recorded directly from the power supply. As shown in Fig. 7(a), larger grid bar spacing increased the effective electrode distance, requiring a higher V_0 and reducing the corona discharge current. This trend aligns with established principles of electrode spacing and corona discharge current [11]. After corona onset, the measured current showed an approximately quadratic increase within the stable voltage range, which is consistent with the shifted-voltage current relation in (6).

Wind speed was measured using an anemometer (405i, Testo Inc.) as shown in Fig. 7(b) and (c). The wind speed initially increases and then decreases as grid spacing widens. At an applied voltage of 11.09 kV, a maximum wind speed of 1.94 m/s is observed at a 5 mm grid spacing, close to the simulated value of 1.90 m/s for the optimized collector geometry. As the voltage approaches the breakdown limit (11.90 kV), the peak wind speed increases to 2.09 m/s. Deviating from this optimal spacing reduces the wind speed. Specifically, narrower spacing exacerbates blockage and pressure loss, while wider spacing diminishes the collected current and compromises focusing. Furthermore, as depicted in Fig. 7(d), the generated thrust was quantified by recording the weight reduction of the IFM on a precision balance (CN-LQBTX5003, Kunshan Ante Inc.). The results in Fig. 7(e) confirm that the 5 mm spacing delivers the highest thrust at voltages above 10.46 kV. While the inherent fluctuations of

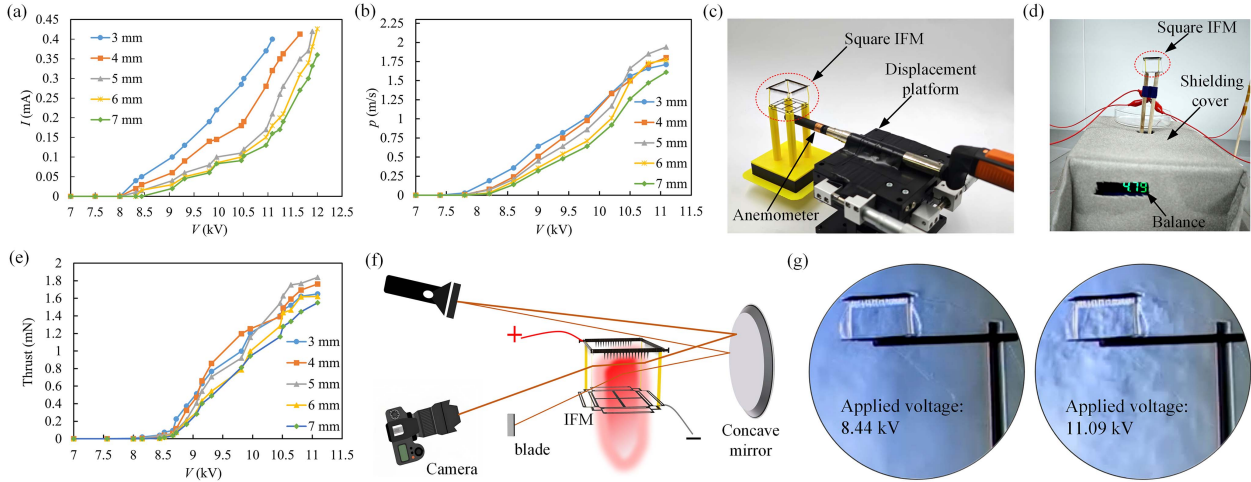


Fig. 7. Measurement and visualization of current, wind speed, and thrust of the square IFM with varying spacings of the additional grid bars. (a) Relationship between current and applied voltage. (b) Relationship between wind speed and applied voltage. (c) Photograph of the wind speed measurement experimental setup. (d) Photograph of the thrust measurement experimental setup. (e) Relationship between thrust and applied voltage. (f) Schematic of the schlieren imaging. (g) Schlieren imaging of the flow field surrounding the square IFM.

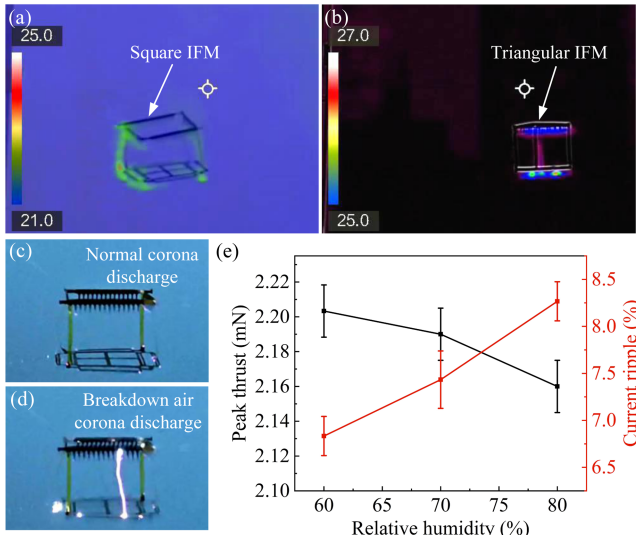


Fig. 8. Characterization of IFM corona discharge. (a) Infrared observation of the square IFM. (b) Infrared observation of the triangular IFM. (c) Photograph of the square IFM normal corona discharge. (d) Photograph of the square IFM under corona discharge breakdown. (e) Humidity-dependent peak thrust and corresponding current ripple.

the corona discharge cause minor thrust variations, the standard deviation remained below 4% of the mean. When the voltage was further increased toward the near-breakdown region, the thrust fluctuation became more pronounced, with the standard deviation increasing to approximately 7.5% near 11.7 kV. For the optimized square IFM at 11.09 kV, the measured current and balance-measured thrust were approximately 0.21 mA and 1.84 mN, respectively. Overall, these findings align with the theoretical predictions of (9) and (11), confirming that an intermediate spacing best balances field focusing with geometric openness.

To visualize the EHD-induced flow field, schlieren imaging was used [see Fig. 7(f)]. The schlieren imaging results of the flow

field surrounding the square IFM reveal that airflow is primarily generated around the lower region of the emitter, with stronger intensity at the central interior [see Fig. 7(g), Supplementary Video 1]. At an applied voltage of 8.44 kV, a distinct airflow appears between the emitter and the collector. When the supply voltage is further increased to 11.09 kV, the ion wind has a higher speed and a larger airflow area. These observations validate simulation results regarding airflow convergence, which directly leads to higher thrust efficiency.

B. Thrust Comparison Experiment

Corona power can be expressed as $P = IV$, and propulsive efficiency η can be defined as the propulsive force generated per unit corona power. According to (3) and (5), it can be obtained

$$\eta = \frac{F}{P} = \frac{d}{\mu V} = \frac{\sqrt[3]{S_{\text{eff}} \cdot \frac{9}{8} \epsilon_0}}{\mu^{2/3}} \cdot \frac{1}{\sqrt[3]{P}}. \quad (12)$$

Equation (12) indicates that for a given gap, a higher voltage increases the thrust but reduces the thrust-to-power ratio. Consequently, maximizing efficiency requires operating just above the corona onset, whereas achieving peak thrust dictates pushing closer to the breakdown limit. The model also establishes a concise design guideline, demonstrating that increasing S_{eff} through collector open-area ratio and field focusing improves efficiency scaling at a constant power. Conversely, merely increasing the input power yields diminishing returns, as the efficiency η decays proportionally to $P^{-1/3}$.

Thrust density is calculated by dividing thrust by the collector's effective open-area projection [25], [26]. For the quad-thruster array [6], the reported $F = 0.295$ mN, $P = 90.40$ mW, and thrust density of 13.67 N/m² give a thrust density per power unit of 151.17 N/(m² · W) and an efficiency of 3.27 mN/W. For RoboFly [27], $F = 0.736$ mN, $P = 60.00$ mW, and an effective wing sweep area of 308 mm² yield a thrust density of 2.39 N/m² and an efficiency of 12.20 mN/W.

For the triangular IFM at the point of takeoff, where F equals the gravitational force of 0.402 mN, a thrust density of 46.21 N/m² is achieved with a collector projected area of 8.7 mm². Given that input power is 0.26 W (8.6 kV and 0.030 mA), the achieved thrust density per power unit escalates to 179.1 N/(m² · W) and the efficiency is 1.56 mN/W. For the square IFM at takeoff, where $F = 0.706$ mN, a thrust density 25.96 N/m² is calculated with a collector projected area of 27.2 mm². The square IFM provides a larger payload-lift margin than the triangular IFM.

These results indicate that IFM efficiency is lower than the flapping-wing RoboFly [27], but the square IFM's efficiency exceeds the quad-thruster array reported in [6] while providing a higher thrust margin for payload lift.

C. Discharge Characterization and Payload Lift

To ensure stable flight, 30 μ m-diameter enameled wires served as both power supply leads and mechanical tethers, secured to the emitter and collector with silver conductive adhesive. Flight experiments compared triangular and square IFMs in thrust, speed, and flight stability. Owing to the high-temperature burn resistance of the CF-PI laminates, the IFM exhibited high reliability. It operated stably under normal corona discharge conditions for more than 6 h without discernible performance degradation. In subsequent testing comprising two additional 6 h cycles, the corona current decreased by less than 3% in each cycle.

Infrared imaging (K20, Hikmicro Inc.) monitored temperature distribution during takeoff [see Supplementary Video 2 and Fig. 8(a) and (b)]. The highest temperature, occurring at the collector power supply connection, was caused by contact resistance between the silver adhesive and the wire but remained below 27 °C. This forestalled wire burnout. Elevated temperatures were also observed at the emitter tips due to prolonged corona discharge. Due to the lower voltage required for flight, the intensity of corona discharge was reduced, resulting in lower maximum temperatures for square IFM tips (~22 °C) compared to triangular IFM tips (~26 °C). This reduction in temperature lowers the risk of electrical breakdown in air and enhances high-voltage stability, thereby effectively mitigating surface oxidation and microcrack formation commonly observed in unprotected electrodes.

Under low-light conditions, a stable corona glow is visible [see Fig. 8(c)]. Breakdown events typically initiate near local geometric discontinuities or near connector surfaces [see Fig. 8(d)]. To reduce this risk, tips are not placed within a 3.7 mm range (the length of tips) around the connectors. This design strategy minimizes the risk of discharge along the connectors, allowing for higher power supply voltage and enhanced EHD force. With the 5 mm grid spacing kept constant, tests were conducted at relative humidity levels of 60%, 70%, and 80%. As shown in Fig. 8(e), as the humidity rose from 60% to 80%, the peak thrust decreased only slightly from 2.20 to 2.16 mN, while the current ripple increased from 6.8% to 8.3%.

Payload-lift tests were performed to evaluate thrust margin under tethered conditions. Representative takeoff frames and

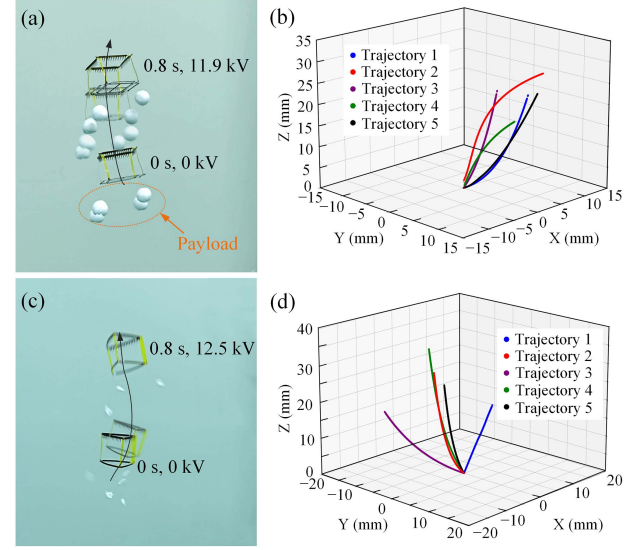


Fig. 9. Flight demonstration of the IFMs with payload. (a) and (b) represent image frames illustrating the takeoff phases and multiple tracked flight trajectories of the square IFM, respectively. (c) and (d) represent image frames illustrating the takeoff phases and multiple tracked flight trajectories of the triangular IFM, respectively.

repeated trajectories are shown in Fig. 9. The assembled masses of the triangular and square IFMs were 41.0 and 72.0 mg, respectively. Across repeated assemblies, the corresponding mass standard deviations were 1.8 and 1.1 mg. The two IFMs lifted payloads of 30.0 and 154.4 mg, respectively, and the square IFM reached a peak thrust-to-weight ratio of 3.14. For five fabricated square IFMs, the propulsion performance was repeatable, with standard deviations of 0.03 mN in peak thrust and 0.05 in thrust-to-weight ratio. The square IFM's larger collector area and stronger flow focusing increase S_{eff} , consistent with its higher payload margin.

The square IFM exhibited less tilt and greater trajectory overlap during repeated flights compared to the triangular IFM. The triangular IFM exhibited a maximum lateral deviation of 10 mm, equivalent to approximately 0.33 body lengths, at an altitude corresponding to its body length. This performance is comparable to that of advanced microrobots reported in previous studies [6], [11], [28], [29]. Increasing the relative humidity from 60% to 72% caused only a minor reduction in the air breakdown voltage for the square IFM, from 11.8 kV to approximately 11.6 kV. However, this reduction in breakdown voltage resulted in a decrease of approximately 10% in its maximum achievable flight altitude.

D. Tethered Closed-Loop Altitude Regulation

Altitude control was achieved using a low-voltage, closed-loop PID architecture. Real-time altitude was measured by a motion capture system (Mars4H, NOKOV Inc.) and streamed to a laptop, where a Python-based PID controller computed the control signal [see Fig. 10(a) and (b)]. To compensate for the slow dynamics of the primary high-voltage supply, the power system employed a dual-channel scheme: a fixed 10 kV bias

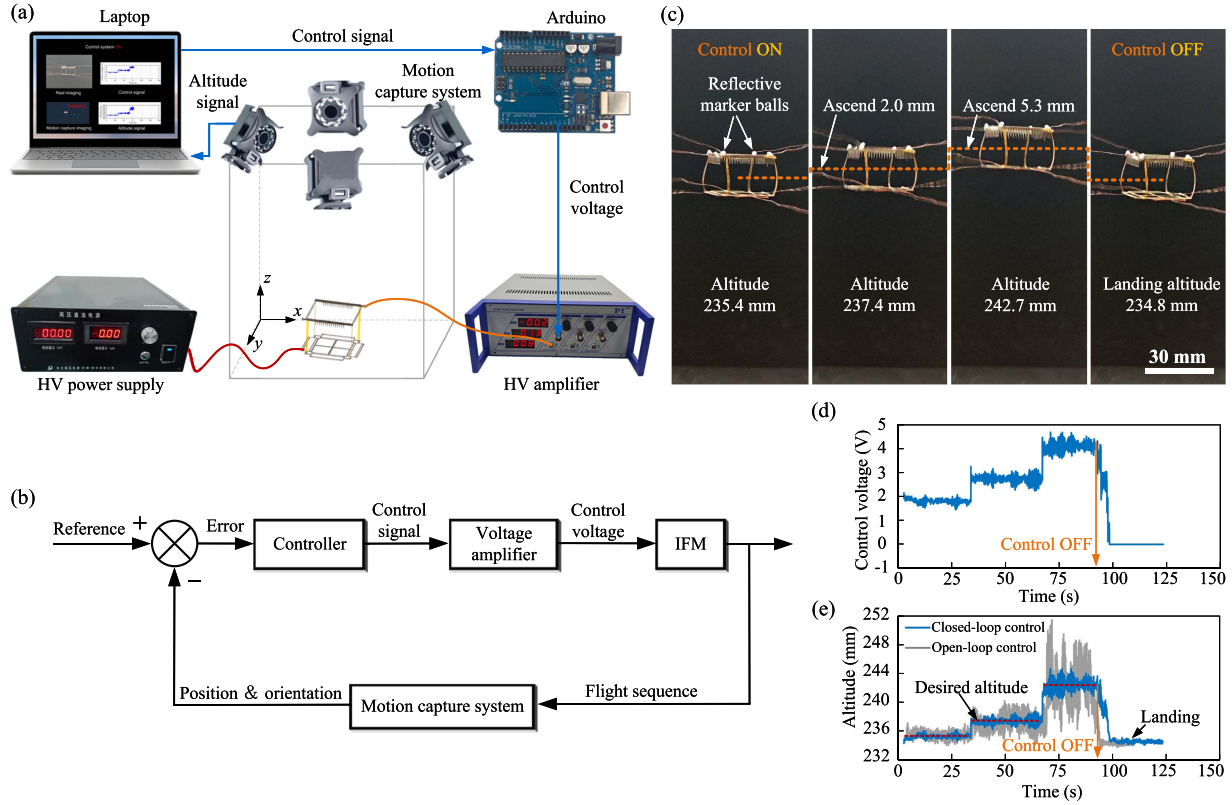


Fig. 10. Flight altitude control experiment of the IFMs. (a) Schematic of the flight altitude control system. (b) Control block diagram. (c) Flight image frames. (d) Control signals generated by the PID controller. (e) The flight altitude collected by the motion capture system.

was applied to the collector electrode to sustain a near-hovering discharge, while a rapid and low-voltage control input modulated the emitter potential through a microcontroller (Arduino Uno R3) and a high-gain amplifier (E-463, Physik Instrumente).

To assess the tether effect, the same 30 μm enameled wires used in the flight experiment were tested separately. Over the 0–6 mm altitude range, the additional downward restoring force exerted by the wires, which opposed the upward motion of the IFM, increased approximately linearly with displacement and reached approximately 88 μN , corresponding to an equivalent tether stiffness of $k_t = 0.0147 \text{ N/m}$. The vertical dynamics of the IFM can therefore be approximated by $m\ddot{z} + c\dot{z} + k_t z = F - mg$, where m is the total mass, z is the altitude, c is an effective damping term, and g is the gravitational acceleration. Around a nominal operating point, the thrust is linearized as $F = F_0 + K\Delta V$, where ΔV is the voltage variation and $K = 0.78 \text{ mN/kV}$ is the fitted thrust gain.

Let $z[k]$ be the measured altitude at discrete time k , and $z_d[k]$ the commanded altitude. The error is $e[k] = z_d[k] - z[k]$. The discrete-time PID law used in the experiments is

$$u[k] = K_p e[k] + K_i T_s \sum_{i=0}^k e[i] + K_d \frac{e[k] - e[k-1]}{T_s} \quad (13)$$

where $u[k]$ is the control voltage applied to the amplifier input, T_s is the sampling interval, and (K_p, K_i, K_d) are tuned gains

($K_p = 0.14$, $K_i = 1.0 \times 10^{-4}$, $K_d = 1.0 \times 10^{-3}$). The amplifier gain is 300, resulting in an emitter modulation range of approximately -1.5 to 0 kV superimposed on the bias condition.

The PID gains were manually tuned and refined to balance overshoot and response time. Experimental results indicate that the proposed control strategy effectively regulates flight altitude. Upon issuing two consecutive ascent commands, the feedback control system adjusted the thrust within 200 ms, yielding sequential altitude increases of 2.0 and 5.3 mm. Descent and landing were executed by reducing the corona voltage via the controller, with the final landing altitude restricted by the bias voltage and tether constraints. Furthermore, the standard deviations of the altitude during the two ascent commands and the landing command were 0.39, 0.82, and 0.15 mm, respectively, as depicted in Fig. 10(c)–(e) (see Supplementary Video 3).

The elevated standard deviation observed at higher altitudes is primarily driven by stronger discharge intermittency at higher operating voltages, as the operating point approaches the breakdown-limited window defined in (8). Nearing breakdown, localized ionization becomes unsteady and transitions toward filamentary discharge. This increases current ripple and, consequently, thrust ripple, thereby degrading altitude tracking performance. This phenomenon is consistent with classical gas discharge physics [10], [11]. In addition, tether compliance and aerodynamic coupling introduce disturbances that become more pronounced as thrust ripple increases.

Across five continuous takeoff, hover, and landing cycles using PID control, the altitude fluctuations within each respective

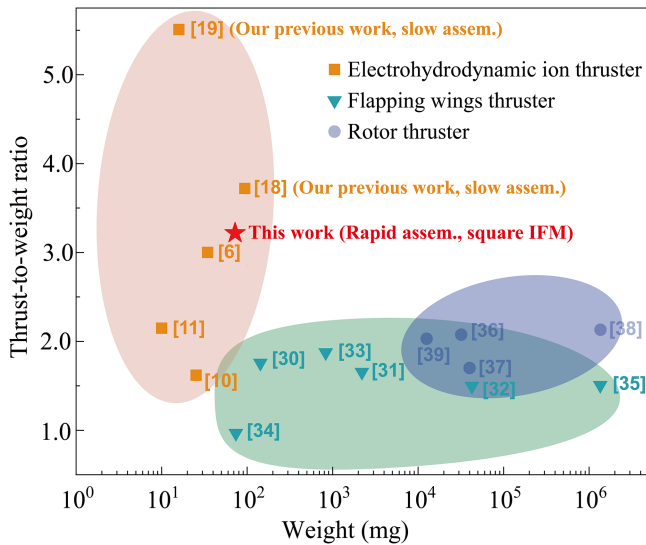


Fig. 11. Comparison of thrust-to-weight ratio and weight in literature reports.

phase remained remarkably consistent. By comparison, open-loop control yielded an altitude standard deviation exceeding 3 mm. Given the open indoor test environment and the finite precision of the motion capture system, an altitude error of less than ± 2 mm is considered acceptable. Ultimately, these findings validate the system's capability to achieve millimeter-scale altitude regulation in a tethered configuration aided by external visual feedback.

Fig. 11 compares the thrust-to-weight ratio and mass of our proposed IFMs with existing literature. The triangular IFM (41.0 mg) and square IFM (72.0 mg) achieve thrust-to-weight ratios of 1.73 and 3.14, respectively. Although lower than our previous aluminized film designs [18], [19], the square IFM slightly exceeds the thrust-to-weight ratios reported for other IFMs [6], [10], [11]. More importantly, the current modules provide a more practical balance between durability and fabrication cost.

Unlike flapping-wing or rotor-based thrusters [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], EHD ion thrusters have no moving parts, reducing mechanical complexity and acoustic signatures. This feature is attractive for carrying tens of milligrams of sensors, such as miniature strain and temperature sensors, in acoustically sensitive monitoring scenarios for several hours. However, untethered operation still requires miniaturized high-voltage conversion and onboard energy storage.

VI. CONCLUSION

This study introduces a novel IFM featuring an origami-inspired structure and CF-PI composites, achieving a balance between lightweight construction, high strength, and efficient assembly. Under tethered conditions, the square IFM demonstrated repeatable vertical takeoff and landing, successfully lifting a 154.4 mg payload (excluding its own weight). This configuration achieved a thrust-to-weight ratio of 3.14, demonstrating its high payload-lift capability. In addition, the internal collector's

convergence effect generated a high wind speed of 2.09 m/s. Infrared imaging and airflow visualizations further elucidated the regional corona discharge and EHD thrust dynamics shaping the flow field, offering critical insights for aerodynamic optimization. Repeated-fabrication tests showed small variations in emitter-collector gap, mass, peak thrust, and thrust-to-weight ratio, and the simulated wind-speed trend was consistent with the measurements.

Despite these promising results, the present work has three main limitations. First, reliance on tethered power and external motion-capture feedback restricts the demonstrated control to vertical altitude regulation rather than free-flight attitude stabilization. Second, increasing payload capacity through higher voltage is constrained by air breakdown, thrust ripple, and structural instability. Third, untethered operation remains limited by the lack of integrated onboard high-voltage conversion and sensing capabilities.

Future work will focus on modular scaling and system autonomy. Larger arrays of low-areal-density CF-PI modules could increase the effective drift area and payload capacity while maintaining manageable voltages. Untethered multi-axis flight will require onboard sensing, such as an inertial measurement unit, for differential thrust management, together with compact HV conversion. Lightweight high-frequency converters and voltage-multiplier topologies, including Dickson or Cockcroft-Walton circuits, are promising under strict isolation and mass constraints. Passive damping frames or compliant stabilizing surfaces may further improve hovering stability if co-designed with the collector geometry and EHD flow path.

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Xianfa Cai (Member, IEEE) received the Ph.D. degree in machine design and theory from Yan-shan University, Qinhuangdao, China, in 2023.

He is currently a Lecturer with the College of Integrated Circuit Science and Engineering, Nanjing University of Posts and Telecommunications, Nanjing, China. His research interests include microrobots and multifields coupled dynamics of microelectromechanical systems.



Yuwang Shen received the B.S. degree in electronic science and technology from the Central South University of Forestry and Technology, Changsha, China, in 2024. He is currently working toward the M.S. degree in microelectronics science and technology with the Nanjing University of Posts and Telecommunications, Nanjing, China.

His research interests include microrobot design, fabrication, and control.



Yingxu Tian received the B.S. degree in electronic science and technology from the North China University of Science and Technology, Tangshan, China, in 2024. She is currently working toward the M.S. degree in microelectronics science and technology with the Nanjing University of Posts and Telecommunications, Nanjing, China.

Her research interests include microrobot design, fabrication, and control.



Yunqi Cao (Senior Member, IEEE) received the B.S. degree in physics from Wuhan University, Wuhan, China, in 2012, the M.S. degree in materials science and engineering from the University of Florida, Gainesville, FL, USA, in 2014, and the Ph.D. degree in electrical engineering from Michigan State University, East Lansing, MI, USA, in 2019.

From 2019 to 2020, he was a Postdoctoral Scholar with the University of Southern California, Los Angeles, CA, USA. He is currently a Tenure-Track Faculty with the College of Control Science and Engineering, Zhejiang University, Hangzhou, China. His research interests include MEMS, flexible electronics, and energy harvesting.



Qiannan Tao (Member, IEEE) received the Ph.D. degree in aerospace propulsion theory and engineering from Beihang University, Beijing, China, in 2024.

She is currently a Lecturer with the College of Integrated Circuit Science and Engineering, Nanjing University of Posts and Telecommunications, Nanjing, China. Her research interests include microaerial vehicle design and robot control methods.



Wei Li (Senior Member, IEEE) received the B.S. degree in mechanical engineering from the Huazhong University of Science and Technology, China, in 2008, and the Ph.D. degree in mechanical engineering from Michigan State University, USA, in 2017.

From 2017 to 2020, he was a Postdoctoral Scholar with the Department of Electrical Engineering and Computer Sciences, University of California, Berkeley, and an Affiliate Postdoctoral Fellow with Lawrence Berkeley National Laboratory, Berkeley, CA, USA. In 2020, he was a Tenure-Track Assistant Professor with the University of Vermont, Burlington, VT, USA. He is currently a Professor with the College of Integrated Circuit Science and Engineering, Nanjing University of Posts and Telecommunications, China, and is also an Adjunct Professor of mechanical engineering with the Huazhong University of Science and Technology, Wuhan, China. His research interests include robotics, MEMS, and mechatronics.